

CHAPTER TWO

LITERATURE REVIEW

2.1 The Human Story of Agricultural Digitalization

When we discuss the "Digitalization of Agriculture," we are navigating a conversation that goes far beyond fancy gadgets, high-speed processors, or complex lines of code. For the smallholder farmer in rural areas, this technology represents a literal lifeline, a bridge between traditional struggle and modern stability. In the context of rural Ghana, digitalization represents a fundamental and necessary shift from a life dictated by uncertainty and "guessing" to a life empowered by informed "knowing" (Addison et al., 2024). Historically, the vocation of farming in West Africa has been a high-stakes gamble against the whims of nature. A single week of delayed rain or a silent pest infestation could mean the difference between a successful harvest and total financial ruin for a family.

Digitalization, as explored by Khaliq et al. (2025), is the strategic integration of modern tools like Artificial Intelligence (AI) and the Internet of Things (IoT) to turn the traditional farm into an "intelligent ecosystem." This ecosystem is designed to be communicative; it allows the land and the livestock to effectively signal their needs to the farmer before a crisis occurs. In our local communities, traditional knowledge has always been the bedrock of farming. For centuries, we have listened to the wisdom of our elders to understand when the rains might arrive or how to treat a wilting leaf using local herbs. However, modern environmental pressures such as rapidly mutating crop diseases and the erratic shifts of a changing climate have made these traditional methods alone insufficient to guarantee survival.

Digitalization acts as the bridge that spans this gap. As Babatunde et al. (2025) note, "Smart Agriculture" is not intended to replace the farmer's hard-earned intuition or their deep connection to the land. Instead, it is designed to sharpen that intuition with data-backed precision. By utilizing digital platforms, we provide farmers with the scientific accuracy they need to protect their livelihoods while allowing them to keep their feet firmly planted in their traditional roots and cultural practices.

2.2 Precision Agriculture: Moving from the Field to the Individual

For decades, the standard for agricultural management was based on a "field-wide" scale. This was a "one-size-fits-all" approach to biology. If one corner of a maize field looked thirsty, the farmer would attempt to irrigate the entire hectare. If a few birds in a coop appeared sluggish, the entire flock was often treated with expensive and sometimes unnecessary antibiotics. This "blanket approach" is not only a massive financial burden on the smallholder, but it is also biologically inefficient and harmful to the local environment. This is where the concept of Precision Agriculture changes the narrative.

Precision agriculture is the practice of observing, measuring, and responding to inter- and intra-field variations in crops and livestock at a very granular level (Babatunde et al., 2025). Sometimes this means focusing down to the health of a single plant or the vital signs of a single bird. According to Khaliq et al. (2025), this transformation is made possible by AI-driven systems that can ingest and analyze environmental data in real-time. Instead of a farmer looking at their farm and seeing a single, indistinguishable green mass, they begin to see it as a collection of thousands of individual lives, each with its own specific health status and nutritional requirements. This level of microscopic detail ensures that vital resources like water, nitrogen, and organic fertilizers are applied only where they are absolutely necessary. This surgical precision is essential for maintaining long-term soil health and ensuring that national food security is built on a sustainable foundation.

2.3 The "Brain" of the Farm: Artificial Intelligence and Machine Learning

In popular culture, Artificial Intelligence (AI) often sounds like a futuristic concept from a science fiction movie. However, in the practical world of our project, AI acts as a "digital extension officer" that lives right in the farmer's pocket. The engine that drives this capability is Machine Learning (ML). To understand the value of ML, we must recognize that unlike traditional computer programs that can only follow pre-set rules, ML allows a system to actually learn from experience and adapt to new data (Sambo et al., 2025).

As Sheshagiri et al. (2025) point out, AI is currently revolutionizing the way we handle agricultural research and extension services. In the past, a farmer would have to wait days or

weeks for an expert to visit their site. Today, when we feed a machine learning model thousands of images of both healthy and diseased crops, the model develops its own mathematical "intuition" for spotting trouble. This is a vital development for Ghana because our human extension officers are often overstretched, sometimes responsible for thousands of farmers across vast districts. AI provides a way to scale this expert-level knowledge, ensuring that every farmer, regardless of how remote their location in the Bono Region may be, has instant access to high-quality, life-saving diagnostic advice (Babatunde et al., 2025).

2.4 Seeing the Invisible: Convolutional Neural Networks (CNN)

If we consider Machine Learning to be the "brain" of the system, then the Convolutional Neural Network (CNN) must be considered the "eye." CNNs are a specific and highly advanced type of deep learning architecture specifically designed to process and interpret visual information. They don't just "see" an image; they perceive it in layers. By breaking a photograph down into tiny mathematical pixels, the CNN detects edges, then identifies textures, then recognizes shapes, until it can identify a complex biological pattern such as the early onset of a fungal disease (Degu & Simegn, 2023).

In the modern world of smart farming, CNNs are now recognized as the gold standard for visual diagnostics in the field. However, a major hurdle in the Ghanaian context is that many rural farmers do not have access to expensive, high-powered computers or high-end smartphones. This is why researchers like Islam et al. (2025) have placed a heavy emphasis on the development of "Lightweight Models." A lightweight model is essentially a "slimmed-down" version of a CNN that has been mathematically optimized to run efficiently on a standard, mid-range smartphone or even through a basic web browser. This means that a farmer standing in a sun-drenched field in Sunyani can take a simple photo of a leaf, and the AI can analyze that image locally on the device without needing a massive data center or a high-speed fiber connection. This accessibility is the core of what makes Agri-Go a truly human-centered and inclusive tool.

2.5 The Poultry Crisis: Fighting Coccidiosis with AI

Poultry farming is one of the most vibrant, fast-moving, and yet terrifyingly vulnerable sectors of the Ghanaian economy. As Sambo et al. (2025) explain, infectious diseases remain the single

greatest cause of economic loss for poultry producers across the continent. One of the most aggressive and persistent enemies a farmer faces is Coccidiosis, a parasitic disease that attacks the digestive tract of birds.

The human tragedy of Coccidiosis lies in its nature as a "hidden" killer. By the time a farmer notices a bird huddling in a corner with ruffled feathers and closed eyes, the parasite has usually already moved through the litter, infecting the entire flock (Degu & Simegn, 2023). For a farmer with a thousand birds, manual inspection of every single animal every day is a physical impossibility. However, the latest research by Islam et al. (2025) shows that lightweight machine learning models can now detect the presence of Coccidiosis with an incredible accuracy rate of over 95%, simply by analyzing digital images of poultry droppings. By identifying microscopic changes in color, moisture, and texture that the human eye would naturally miss, the AI provides the farmer with a vital head start. This "early warning" allows them to isolate the sick birds immediately, treat the flock early, and save their entire life's investment from being wiped out.

2.6 Breaking the Middleman Monopoly with E-Agri Commerce

In the life of a farmer, growing a healthy, bountiful crop is a massive technical and physical triumph. However, the second half of the battle selling that produce for a price that rewards their hard work is often the most difficult part. In the traditional Ghanaian agricultural market, the "middleman" often holds an unfair amount of power. Because farmers in rural areas often lack real-time information about what prices are being paid in large cities like Accra or Kumasi, they are frequently forced to sell their produce for a mere fraction of its true market value (Muniraju & Neelakanta, 2024).

E-Agri Commerce is the digital answer to this economic inequality. It involves using digital platforms to bypass traditional barriers and connect the farmer directly to the final buyer. As noted by Addison et al. (2024), this digitalization of the marketplace is one of the most effective and direct ways to improve rural livelihoods. When a farmer uses an integrated platform like Agri-Go to list their maize, vegetables, or eggs, they are no longer at the mercy of whoever happens to drive past their farm gate that day. They gain "wider reach," which allows them to negotiate with customers in distant urban centers while they remain on their land. This direct

connection facilitates "higher profits" for the producer and leads to "reduced waste," because produce finds a confirmed buyer before it ever has the chance to rot in the sun.

2.7 The Social Heart of the System: Extension and Community

It is crucial to remember that technology is only truly valuable if it improves the quality of human life. Digitalization in agriculture serves a much higher social purpose than just increasing efficiency; it is a tool for strengthening national food security. When our farmers are successful, stable, and profitable, our local markets become full of affordable, healthy, and high-quality food (Sheshagiri et al., 2025).

Furthermore, these digital tools act as a "virtual bridge" for the exchange of information. In many regions of Ghana, the ratio of agricultural extension officers to farmers is dangerously and frustratingly low. By integrating an AI-powered chatbot into a system like Agri-Go, we are effectively providing an "extension officer in a pocket." This system doesn't just diagnose; it educates. It can answer urgent questions about fertilizer mixing, pest control safety, or new government regulations instantly (Babatunde et al., 2025). This empowers the farmer to become more independent, knowledgeable and resilient, significantly reducing the time they spend waiting for external help that in many cases, might never arrive.

2.8 Overcoming the Barriers: Designing for the Real World

Even with the overwhelming evidence of these benefits, we must remain honest about the challenges that exist on the ground. Many farmers are understandably hesitant to adopt new, unproven technology. There is a significant "digital divide" in rural Ghana, fueled by inconsistent internet coverage, a lack of reliable electricity, and a general, deep-seated lack of trust in "machines" and "algorithms" (Addison et al., 2024).

To win the trust of these farmers, the system must be designed for their specific reality. This is what we call "Human-First" design. The user interface of an agricultural app shouldn't look like a complicated computer science textbook; it should use clear, recognizable icons, localized context, and simple, conversational language. By focusing on "Lightweight Models" that are specifically built to function on slow, 2G or 3G connections (Islam et al., 2025) and creating

interfaces that feel as comfortable as a "helpful neighbor," we can ensure that this digital revolution is inclusive. It must benefit everyone in the village, not just those with high-end devices in the big cities.

2.9 Summary of the Research Gap

In conclusion, while many global researchers have explored the potential of Artificial Intelligence for agriculture in controlled laboratory settings, very few have successfully combined high-accuracy AI diagnostics with a functional commercial marketplace in a way that is truly accessible to the Ghanaian smallholder. Most existing agricultural tools are either too technically complex for the average user or require high-speed internet that our rural communities simply do not possess. This project, Agri-Go, aims to close that specific gap. It provides a lightweight, integrated, and locally-optimized solution that is designed to protect both the biological health of the farm and the financial wealth of the farmer, ensuring a sustainable future for Ghanaian agriculture.